Where You Screen Decides What You Conclude: Instance Space Analysis for Quantum-Inspired Resource Allocation in Non-Terrestrial Networks

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Abstract. Quantum-inspired (QI) metaheuristics are widely proposed for resource allocation in space-air-ground integrated networks, and the evidence offered for them is usually a comparison on a handful of problem configurations, against a small set of population-based baselines, over few repetitions. We ask what changes when three evaluation conventions that are already standard in neighbouring communities are applied together to one concrete non-terrestrial problem: mapping the instance space rather than sampling one point in it, giving every method the same function-evaluation budget with restarts permitted, and reporting a method effect against the spread the random seed alone produces. On a multi-beam power allocation problem we map 48 configurations with 3 instances each and compare four optimisers over 30 seeds at a matched budget of 20 000 evaluations, for 144 units in total. Four results follow. First, 24 of 48 configurations are multimodal, and multimodality is driven almost entirely by amplifier distortion rather than by interference or beam count, rising from 17% to 100% of units across that one knob. Second, across all four optimisers restart local search is best or tied in 144 of 144 units, ahead of the QI method (65), CMA-ES (50) and differential evolution (27); and against the usual argument for population-based search, the QI method ties restart local search on 81% of unimodal units while losing on 98% of multimodal ones. Third, every effect we measure is smaller than the seed-induced spread while its direction never once reverses across 144 units. Fourth, reconstructing 5 000 studies at the size typically reported, we find the headline is decided by two choices of the experimenter rather than by the problem: against differential evolution 97.3% of such studies report the QI method ahead, against restart local search 89.4% report it behind, and among studies that happen to screen only unimodal configurations 67.4% report no difference at all. We release the map and the harness.

Keywords: Quantum-inspired optimisation · Non-terrestrial networks · Instance space analysis · Benchmarking · Reproducibility · Resource allocation.

1 Introduction

Section 2 gives the setting, Section 3 separates the conventions we adopt from the observations we contribute, and Section 4 states the protocol; the implications are drawn together in Section 6.

Space-air-ground integrated networks (SAGINs) combine low Earth orbit constellations, high-altitude platforms, unmanned aerial vehicles and terrestrial cells into one programmable fabric, and non-terrestrial networks (NTN) are now a first-class component of the 3GPP releases. The control problems these systems pose share a shape: beam steering, handover, slice placement, trajectory planning and spectrum assignment all produce decision spaces that grow rapidly with the number of nodes, and all must be solved under severe energy and latency budgets. That shape is an invitation to metaheuristics, and quantum-inspired (QI) methods are among the families most frequently proposed for it, on the argument that they bring a complementary search behaviour while running on today's classical hardware.

The question this paper raises is not whether such methods can help. It is narrower and, we believe, more urgent for a community still forming its conventions: *what is a reader entitled to conclude from the evidence that is typically presented?*

That evidence has a recognisable shape. A QI optimiser is run on a few problem configurations, compared against one or two population-based baselines such as a genetic algorithm or differential evolution, and reported over few repetitions. Each of those three choices has a known failure mode, and each has an established remedy in a neighbouring community: instance space analysis in algorithm selection, restart-fair protocols in optimisation benchmarking, and seed-variance reporting in machine learning reproducibility. The contribution of this paper is not to invent those remedies. It is to apply them together, to one concrete non-terrestrial allocation problem, and to report what changes.

What we do. Figure 1 summarises the design. We take a multi-beam power allocation problem with amplifier distortion and run two arms under one protocol. The first maps the *instance space*: for every configuration in a grid we ask whether the objective is multimodal, using multi-start local search with an explicit separation threshold and with three categories of run excluded from the count and reported rather than silently dropped. The second compares four optimisers at a matched evaluation budget with restarts permitted, repeated over 30 seeds, with the unit of analysis declared in advance. Crossing the two arms lets us ask a question neither answers alone: does position in the instance space predict the verdict? Finally we use the full data to reconstruct what a study of the size usually reported would have concluded.

What we find. Position does predict the verdict, and in the direction opposite to the usual argument for population-based search. The QI method is competitive precisely where the landscape is *unimodal*, and is beaten almost everywhere it is multimodal, because multimodality is also what makes cheap restarts effective.

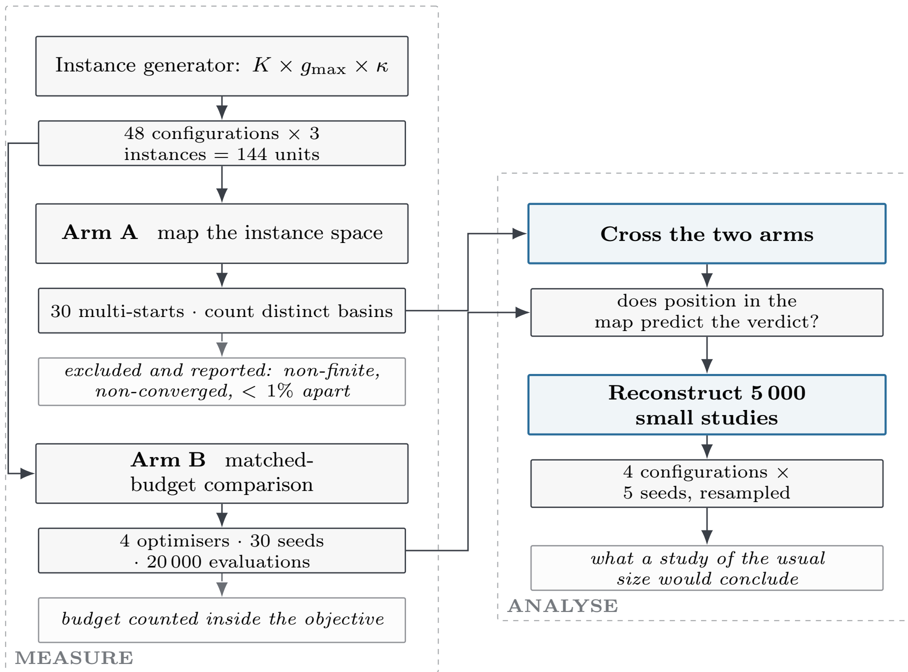


Fig. 1. The protocol. Two measurement arms run over the same grid of units; crossing them asks whether position in the map predicts the verdict, and resampling the full data asks what a study of the size usually reported would have concluded.

More practically, the reconstruction shows that the headline a small study reports is governed by two choices the experimenter makes, the baseline and the screening location, more than by the problem itself.

Why this belongs in a workshop on QI-AI for SAGINs. None of the three conventions is new. What is new, to the best of our knowledge, is their joint application to a telecommunications resource allocation problem, the resulting map, and the quantified demonstration that conclusions in this area are sensitive to design choices that are rarely reported. For a community assembling its evaluation practice, that is more useful than another method would be.

2 Background

2.1 Resource allocation in non-terrestrial networks

Power and beam allocation in multi-beam satellite and aerial systems is a long-standing problem [9]. The objective is typically a sum rate or a fairness-weighted variant, the constraints are a total power budget and per-beam limits, and the coupling between beams comes through interference; formulations of this kind appear both as fractional programmes and as joint beam-direction and resource problems in dynamic LEO constellations [11]. Non-terrestrial access became a first-class part of the standards with 3GPP Release 17, and the deployment experience since then is surveyed in [10]. Two features make the non-terrestrial case harder than its terrestrial counterpart.

The first is mobility. The geometry changes continuously, so an allocation is solved repeatedly rather than once, and the operating point moves through the parameter space over the course of a pass. The second is the payload. On-board amplifiers are driven close to saturation to conserve power, and the resulting nonlinearity introduces a term that is neither convex nor separable.

Both features matter for evaluation, and the first matters most for this paper. If the operating point moves, then a claim established at one operating point is not automatically a claim about the system. This is why learning and optimisation methods for these networks are increasingly evaluated over a range of operating conditions rather than a nominal one [9]. This is the sense in which evaluation for NTN is evaluation under non-stationarity, and it is why a map is the right object to report rather than a point.

2.2 Quantum-inspired metaheuristics

QI metaheuristics adapt notions from quantum mechanics, among them superposition, amplitude encoding and rotation operators, into classical search procedures. Quantum-inspired particle swarm optimisation replaces the velocity update of standard particle swarm with a sampling step drawn from a distribution whose width contracts over the run. Quantum-inspired evolutionary algorithms represent a population as probability amplitudes and update them with rotation gates. These methods run entirely on classical hardware, which is the source of their appeal for on-board or edge deployment where no quantum device is available. They have been applied to UAV trajectory planning [12] and, more recently, to trajectory optimisation in low-altitude networks [13]; broader assessments of quantum and quantum-inspired optimisation for wireless systems are given in [14,15], both of which conclude that these methods are best regarded as promising heuristics rather than established outperformers of classical solvers.

A separate line of work has examined the search operators of such methods directly and released a benchmark harness for that purpose [16]. That work studies synthetic benchmark functions and asks what the QI operators *are*. The present paper studies one real allocation problem and asks what a practitioner may conclude from a typical evaluation of them. The two are complementary and we do not restate its findings here.

3 Three conventions we adopt but do not claim

We separate carefully the conventions we adopt, which are not ours, from the observations we contribute, which are.

3.1 Budget-matched comparison with restarts permitted

Comparing stochastic optimisers under a common budget, with restarts allowed rather than forbidden, has been formalised as a restart-fair benchmarking protocol [1], and restart schedules are an established baseline in their own right [2]. Best practice for optimisation benchmarking more broadly is surveyed in [3].

Why this matters here is specific. A population method given a fixed budget and compared against a local search that is *not* permitted to restart is being compared against a handicapped opponent, because a local search that has converged has nothing left to do with its remaining budget. Permitting restarts is not a courtesy to the baseline; it is what makes the budget meaningful.

3.2 Instance space analysis

That algorithm performance varies systematically across a structured space of problem instances, and that a footprint in that space rather than an aggregate is the right object to report, is the instance space analysis (ISA) methodology [4,5]. ISA has been applied to classification, regression, scheduling, knapsack and maximum flow, and recently to the instance dependence of optimal parameters in a quantum optimisation algorithm [6]. To the best of our knowledge it has not been applied to a telecommunications resource allocation problem.

3.3 Seed variance as the yardstick for an effect

That the random seed alone can move a reported result by more than the method under study is well documented in machine learning benchmarking [7,8], with single-seed reporting shown to be a poor estimator of the repeated-run mean [8]. The practice we adopt is to report an effect *relative* to that spread, so a reader can see whether a difference is larger or smaller than the noise floor of the experiment itself.

4 Problem and protocol

4.1 The allocation problem

We allocate transmit power $p \in \mathbb{R}^K_{\geq 0}$ across K beams under a total power constraint $\sum_k p_k \leq P_{\text{tot}}$, maximising the sum rate

$$R(p) \;=\; \sum_{k=1}^{K} \log_2\!\left(1 + \frac{g_{kk}\, p_k}{\sigma^2 + \sum_{j \neq k} g_{kj} p_j + \kappa\, (\mathbf{G}p^3)_k}\right), \tag{1}$$

where $\mathbf{G}$ collects channel and interference gains, σ^2 is the noise power, and κ scales a cubic term standing for amplifier nonlinearity. Diagonal entries g_{kk} are drawn uniformly from $[1, 2]$ and off-diagonal entries uniformly from $[0, g_{\max}]$, so $g_{\max}$ controls interference strength. We set $P_{\text{tot}} = K$ and $\sigma^2 = 1$.

Three knobs generate the instance space: the beam count K, the interference bound $g_{\max}$, and the distortion strength κ. Writing $\mathcal{C}$ for the configuration grid and $\mathcal{U}$ for the set of units,

$$\mathcal{C} \;=\; \mathcal{K} \times \mathcal{G} \times \mathcal{X}, \qquad \mathcal{U} \;=\; \{\, u = (c, i) \;:\; c \in \mathcal{C},\; i = 1, \ldots, I \,\}, \tag{2}$$

with $|\mathcal{C}| = 48$, $I = 3$ instances per configuration and $|\mathcal{U}| = 144$. The unit u, not the individual run, is the unit of analysis; seeds are repetitions *inside* a unit.

4.2 Arm A: mapping the instance space

For each unit of (2) we run 30 independent local searches from uniformly drawn starting points and count how many *distinct* local optima they reach. Three categories are excluded from that count, because including any of them would manufacture multimodality:

1. runs whose objective became non-finite, since the solver then stops wherever it happens to be and that point is not a local optimum;
2. runs that did not converge, having reached the iteration limit or terminated abnormally in the line search;
3. optima separated by less than 1% of the best sum rate, which is the local solver's own convergence noise rather than a materially different basin.

Formally, let $V(u) = \{v_1 \geq v_2 \geq \cdots \geq v_m\}$ be the objective values of the scored runs at unit u, sorted descending. The retained set $B(u)$ is built greedily,

$$B(u) \;=\; \{v_1\} \cup \Big\{\, v_j \;:\; \min_{w \in B(u)} |v_j - w| \;>\; \tau\,|v_1| \,\Big\}, \qquad \tau = 0.01, \tag{3}$$

and the unit is called multimodal when more than one basin survives,

$$\mathrm{MM}(u) \;=\; \mathbf{1}\big[\, |B(u)| > 1 \,\big]. \tag{4}$$

The excluded counts are recorded rather than discarded, so a reader can see how much of the sample each rule removed. The threshold τ is a choice, and we return to it in Section 7.

4.3 Arm B: budget-matched comparison

Four optimisers are compared: a quantum-inspired particle swarm method, CMA-ES, differential evolution and random-restart L-BFGS-B. Every method receives the same budget of 20 000 objective evaluations, and the budget is *counted inside the objective* rather than estimated from iteration counts, so it is enforced rather than assumed. Each method is repeated over $S = 30$ seeds. The budget is enforced by a counter carried in the objective itself: writing $n_m(u, s)$ for the number of calls a method m has made,

$$\text{a run terminates as soon as } \; n_m(u, s) \;\geq\; N_{\max} = 20\,000, \tag{5}$$

so that no method can exceed the budget through an implementation detail such as an inner line search. For each unit and method we record the mean and the spread across seeds,

$$\mu_m(u) = \frac{1}{S}\sum_{s=1}^{S} f_m^\star(u,s), \qquad \sigma_m(u) = \Big[\tfrac{1}{S-1}\textstyle\sum_s \big(f_m^\star(u,s) - \mu_m(u)\big)^2\Big]^{1/2}, \quad (6)$$

where $f_m^\star(u,s)$ is the best sum rate method m reached at unit u with seed s.

The quantity we report for a pair of methods (a,b) is the effect measured against the spread the seed alone produces,

$$\Delta_{ab}(u) = \mu_a(u) - \mu_b(u), \qquad r_{ab}(u) = \frac{|\Delta_{ab}(u)|}{\max\{\sigma_a(u),\, \sigma_b(u)\}}, \quad (7)$$

with r_{ab} defined only where the denominator is positive. Units where it vanishes are not discarded: both methods were deterministic across all S seeds there, which is itself reported in Section 5.

4.4 Reconstructing an underpowered study

The full data set permits a question no single experiment can answer: what would a study of the size usually reported have concluded? We draw 5 000 synthetic studies. Each selects 4 configurations at random, one instance within each, and 5 of the 30 available seeds, then reports its verdict by majority across its units.

Formally a study draws $\mathcal{C}_0 \subset \mathcal{C}$ with $|\mathcal{C}_0| = 4$, one instance per selected configuration giving units $\mathcal{U}_0$, and a seed subset Σ_0 with $|\Sigma_0| = 5$. Writing $\hat\mu_m(u)$ for the mean over Σ_0 only, the study reports

$$V(\mathcal{U}_0) \;=\; \operatorname{sign} \sum_{u\in\mathcal{U}_0} \operatorname{sign}\big(\hat\mu_a(u) - \hat\mu_b(u)\big), \quad (8)$$

that is, a majority verdict across its own units. This is resampling, not new measurement. It says how well a small design resolves the effect, *given* that the full data describe the truth. It is a statement about the design, not about the world, and it creates no information the full data do not already contain.

5 Results

5.1 A single screening point can report the opposite of the map

Figure 2 shows the instance space. Under the criterion of (3) and (4), 24 of the 48 configurations contain at least one multimodal instance, so multimodality is common but far from universal.

The highlighted tile matters more than the rest. It is the configuration at which an earlier screening run for this problem was carried out, at $K = 8$, $g_{\max} = 0{,}30$ and $\kappa = 0{,}05$. In our map that configuration is unimodal in 3 of 3 instances. A screening run there finds no spread of local optima and invites the conclusion that the problem class has nothing for a global search method to find. The map shows that conclusion holds for that configuration, not for the class.

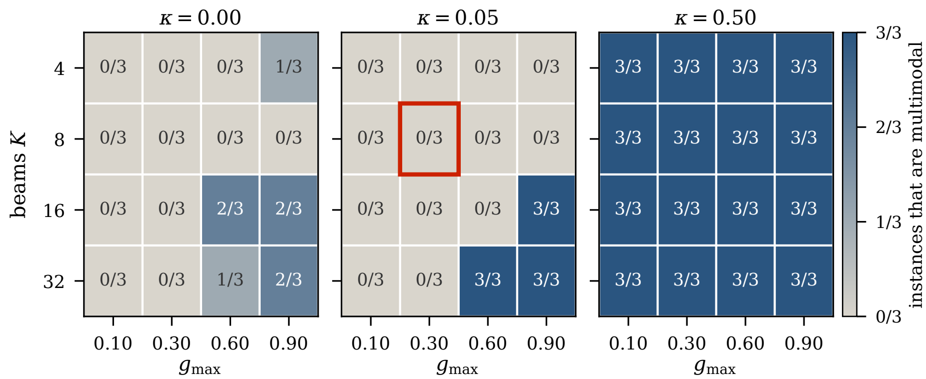


Fig. 2. Instance space of the allocation problem. Each tile reports how many of the 3 instances at that configuration are multimodal. The outlined tile is the single configuration at which an earlier screening run was performed.

5.2 Multimodality comes from amplifier distortion, not from interference

Figure 3 breaks the map down by knob, and the structure is not what one would guess. Beam count and interference strength both matter, but weakly: the multimodal fraction rises from 33% to 58% across the range of K, and from 33% to 64% across $g_{\max}$. The distortion term is different in kind. It moves the fraction from 17% to 100%, and the movement is a step rather than a gradient: the two lower levels behave alike and the highest level is multimodal everywhere.

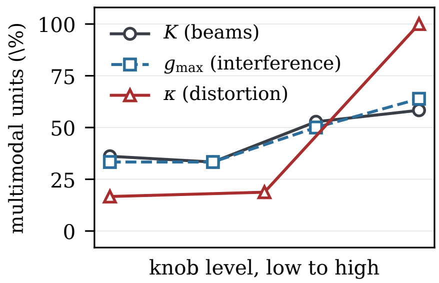
This has a practical consequence for experiment design in this area. Interference strength is the knob most often swept, because it is the one with an obvious physical interpretation for a multi-beam system. It is not the knob that decides whether the optimisation problem is hard. A sweep over $g_{\max}$ at low distortion explores a region that is overwhelmingly unimodal, and a global search method evaluated there has nothing to demonstrate.

5.3 All four optimisers, before splitting by region

Table 1 reports the four optimisers over the whole grid, before any split by region, ranking them by the unit means $\mu_m(u)$ of (6) and reporting the seed spread $\sigma_m(u)$ alongside. Restart local search is best or tied in every unit. The two population baselines sit below the QI method on mean rank, which is the comparison a paper in this area would typically report and would typically report as a win.

5.4 Position in the map decides the verdict

Table 2 and Fig. 4 cross the two arms. Restart local search is best or tied in 144 of 144 units, and the QI method wins 0 of them. That alone would be a familiar negative result. The structure behind it is the part worth reporting.



knob	level	units	multimodal (%)
K	4	36	36
	8	36	33
	16	36	53
	32	36	58
$g_{\max}$	0.1	36	33
	0.3	36	33
	0.6	36	50
	0.9	36	64
κ	0	48	17
	0.05	48	19
	0.5	48	100

Fig. 3. Fraction of units that are multimodal, by knob and level. Beam count and interference move the fraction gradually; distortion moves it as a step, from 17% at the two lower levels to 100% at the highest.

Table 1. All four optimisers over all 144 units. Mean rank is averaged over units, lower being better. The last column is the median standard deviation across the 30 seeds.

method	best or tied	mean rank	median seed s.d.
QI particle swarm	65/144	2.25	0.0000
CMA-ES	50/144	2.50	0.0059
Differential evolution	27/144	3.56	0.0007
Restart L-BFGS-B	144/144	1.69	0.0000

On unimodal units the QI method *ties* restart local search in 81% of cases: both find the same solution and the QI machinery buys nothing, because there is one basin and any competent method reaches it. On multimodal units it *loses* in 98% of cases.

This inverts the argument usually given for population-based search. Diversity is supposed to pay on rugged landscapes. Here multimodality is not where diversity pays; it is where cheap restarts pay. The reason is not mysterious. Sampling many basins independently is exactly what a restarted local search does, and it does so at the cost of one convergence per basin, while a population spends its budget maintaining a diversity it cannot convert into basin coverage at the same rate.

5.5 Every effect is smaller than the seed spread, and no direction reverses

The right panel of Fig. 4 groups the units by the ratio r_{ab} of (7), which places each method difference against the spread caused by the seed alone. The median ratio is 0.27 on unimodal units and 0.70 on multimodal units. Both are below

Table 2. QI method against restart local search, split by region of the instance space. The last column is the absolute mean difference divided by the seed-induced standard deviation.

region	units	QI wins	tie	loss	effect / seed spread
unimodal	79	0	64	15	0.27
multimodal	65	0	1	64	0.70

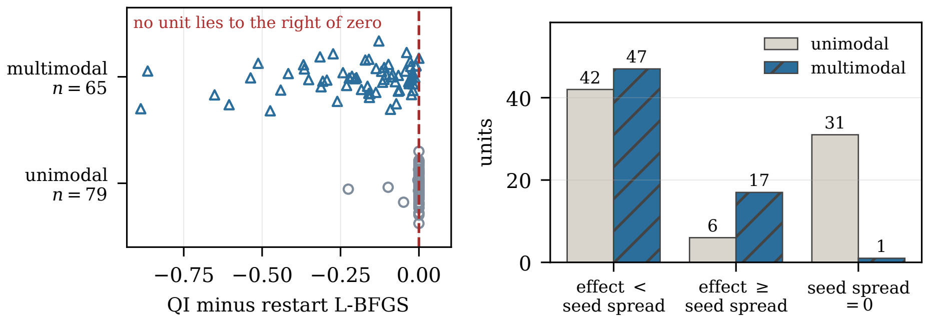


Fig. 4. Left: every one of the 144 units, plotted by the difference in mean sum rate between the QI method and restart local search; no unit falls to the right of zero. Right: the same units placed by effect size against seed spread, with the diagonal marking equality. The 32 units in which both methods were deterministic across all 30 seeds are drawn on the lower band rather than dropped.

one: the difference between the two methods is smaller than the difference one observes by changing the random seed and nothing else.

It would be wrong to read this as "there is no effect". The direction is perfectly consistent: across all 144 units the QI method wins 0. An ordering this stable but this small is exactly the regime in which an underpowered study returns whichever answer its seeds happened to give, and in which a single-seed study cannot be distinguished from a study reporting noise. The next subsection quantifies that.

5.6 The baseline is also the reproducible one

A result that emerged only when we stopped discarding the zero-variance units deserves separate mention. Restart local search returns the same solution on every one of the 30 seeds in 95% of units (137 of 144). The corresponding figures are 35% for CMA-ES, 22% for the QI method and 15% for differential evolution.

For a deployment context this matters independently of solution quality. An allocation recomputed on every pass, on a payload with no operator in the loop, is easier to certify and to debug when it does not depend on a seed. The baseline that wins on quality here also happens to be the one whose output is reproducible, and a comparison that reports only mean sum rate does not show this.

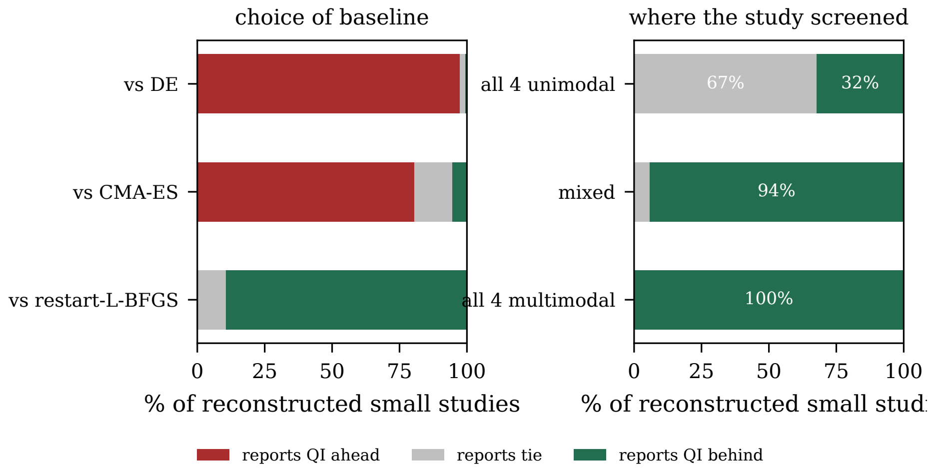


Fig. 5. Verdicts of 5 000 reconstructed studies at the size usually reported. Left: the verdict as a function of which baseline the study chose. Right: the verdict as a function of where in the instance space the study happened to screen.

We note that this is also why the right panel of Fig. 4 would have been misleading had we drawn it naively. A ratio of effect to seed spread is undefined when the spread is zero, and dropping those 32 units would have removed from the figure precisely the units where the baseline is most dependable.

5.7 What a study of the usual size would have concluded

Figure 5 reports 5 000 reconstructed studies of 4 configurations and 5 seeds each, each reporting the majority verdict of (8). Two findings follow, and the second is the one we would ask readers to take away.

The baseline decides the headline. Against differential evolution, 97.3% of small studies report the QI method ahead. Against CMA-ES, 80.5% do. Against restart local search, 89.4% report it behind. All three are faithful to the full data, which show the QI method ahead of both population baselines and behind restart local search. No error is involved and no bad faith is required: a paper that compares against a genetic algorithm and a differential evolution baseline and reports a win is reporting something true. It is simply not reporting the comparison that decides whether the method is worth deploying.

Where the study screened decides whether it sees anything at all. Among reconstructed studies whose 4 configurations all fell in the unimodal region (411 of 5 000), 67.4% report no difference between the QI method and restart local search. Among those whose configurations all fell in the multimodal region (174 of 5 000), 100.0% report the QI method behind. Same problem, same methods, same budget, opposite headline, decided by a sampling choice that such papers rarely state and never justify.

In fairness to the small design. Against restart local search the small design does mostly reach the correct direction, with only 0.1% of studies reporting the QI method ahead. The failure mode here is not that small studies invert the truth. It is that they resolve it into "no difference" 10.5% of the time overall and 67.4% of the time when they screen in the easy region, and a reported non-difference is precisely what allows a method to remain in circulation unexamined.

6 Discussion

6.1 Three things a paper in this area should report

The results suggest three additions to reporting practice, all of them cheap.

State where you screened, and screen more than one point. The map in Fig. 2 took a few minutes of computation, far less than the optimiser comparison it contextualises. A paper that reports the configurations it swept, and whether they are multimodal under a stated criterion, gives a reader what is needed to judge whether the comparison had anything to detect.

Include a restarted local search at matched budget. It is the baseline a reviewer reaches for first, it is trivial to implement, and in this problem it is the only baseline that changes the conclusion. A comparison against population methods alone cannot distinguish "this QI method searches well" from "this problem does not need a population".

Report the effect against the seed spread. A difference of a given size means something different in a problem where seeds move the result by half that much than in one where they move it by twice as much. The ratio costs nothing once repetitions exist, and it tells a reader immediately whether the study had the resolution to see what it claims to have seen.

6.2 What this does not say

It does not say QI methods are useless for non-terrestrial resource allocation. It says that on this problem, under this protocol, one QI method does not beat a restarted local search, and that the published evidence pattern in this area would frequently not have detected that. A method genuinely better would survive the same protocol, and the protocol would then be evidence in its favour rather than an obstacle. That is the sense in which the contribution is intended as constructive rather than negative.

Nor does it say small studies are unreliable in general. The reconstruction shows the opposite in one respect: where an effect is consistent, a small design mostly recovers its direction. What the small design loses is not the sign but the resolution, and with it the ability to tell a real non-difference from an unexamined one.

6.3 A note on generalisation

The mechanism behind our central finding is not specific to this objective. Wherever multimodality arises from a smooth nonlinearity on a boxed domain, basins are cheap to sample and a restarted local search will be a strong baseline. We would expect the finding to weaken where basins are expensive to reach, for example under integrality constraints or where evaluating the objective requires a simulation rather than a closed form. That expectation is testable, and mapping the instance space of such a problem is the way to test it.

7 Threats to validity

One problem family, one generator. The map is of one allocation problem under one instance generator. Nothing here establishes how QI methods behave on scheduling, routing or placement problems in the same networks, and the instance space of those problems would have to be mapped separately. What transfers is the protocol; the verdict does not.

The multimodality criterion is a choice. The threshold τ of (3) treats optima separated by less than 1% of the best sum rate as one basin. A stricter threshold would report more multimodality and a looser one less. The threshold is stated rather than implied, and the excluded non-finite and non-converged runs are counted rather than dropped silently, so a reader can see the size of each exclusion.

Restart local search is a baseline, not a recommendation. That it wins here follows from the geometry of the objective in (1), which is smooth and box-constrained. It is the baseline a reviewer reaches for first, which is why it is the one we ran. It is not a claim about what should be deployed on a payload, where memory and code size constraints may exclude it.

Budget is counted in evaluations, not wall-clock time. The stopping rule of (5) is the harder constraint to satisfy and the easier one to verify, but it treats generously a method whose per-call overhead is large. We report the evaluation budget rather than a timing comparison for that reason, and a wall-clock study would be a useful complement.

The reconstruction is resampling, not replication. Section 5 reports what a small design would conclude *given* that the full data describe the truth. It cannot speak to a small study run on a different generator, and it creates no information the full data do not already contain. It is a statement about the resolving power of a design.

One QI variant. We ran one quantum-inspired particle swarm method. Other QI families, in particular quantum-inspired evolutionary algorithms with rotation-gate updates, may behave differently, and this paper makes no claim about them.

8 Conclusion

Applying three established evaluation conventions together to one non-terrestrial allocation problem changes what the evidence supports. A screening run at a single configuration can land in the unimodal fraction of the instance space and conclude the opposite of what the map shows; multimodality in this problem is created by amplifier distortion rather than by the interference term that is usually swept; a budget-matched restart baseline is best or tied in 144 of 144 units; and every effect we measure is smaller than the spread the random seed alone produces, while its direction never once reverses.

Reconstructing 5 000 studies at the size usually reported shows why this matters in practice. The headline such a study produces is governed by the baseline it chose and the region it happened to sample, both decisions of the experimenter, neither customarily reported. The map and the harness are released so the screening step can be run before a claim is made rather than after it is disputed.

Reproducibility. The instance space map, the raw per-seed results for all 144 units, the reconstruction, and the harness that produced them are released with the paper. Every number in this paper is generated from the released result files by a single command; no result value is typed into the manuscript source.

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